



TOSHKENT SHAHRIDAGI INHA UNIVERSITETI

INHA UNIVERSITY IN TASHKENT

STUDENT NAME:

STUDENT ID NUMBER:

SECTION NUMBER IN THIS COURSE:

1 / 2 / 3

FALL – 2024 (Mid-Tem Examination)

COURSE NAME:

Software Engineering

COURSE NUMBER:

CSE4070

EXAMINATION DATE:

TIME:

EXAMINATION DURATION:

60 minutes

ADDITIONAL MATERIALS
ALLOWED TO USE:

Not allowed.

SPECIAL INSTRUCTIONS:

All questions compulsory.
Total Marks 25

Please do not open the examination paper until directed to do so.

READ INSTRUCTIONS FIRST:

Desks should be free from all unnecessary items (books, notes, technology, food, water, clothes)

Use of any electronic device (Phone, iPod, iPad, laptop) is not allowed during the examination.

Cheating, talking to fellow students, singing, turning back are not allowed

Write your Name (*capital letters*), ID number and Section number in each page of your examination paper.

Final answers must be written by **only blue or black, non-erasable pen**. Do not use highlighters or correction pen.

All answers should be written in the space provided for each question, unless specified the other way.

You should choose one answer in the multiple-choice question.

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If you have a problem, please raise your hand and wait quietly for a Proctor.

You are not allowed to leave the exam room until you submit the exam papers.

Sample

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Section A - Answer the following Questions (7 X 2 = 14 Marks)

1. What is a Legacy System? Why should it be replaced?

2. Explain the Waterfall Software Development Process. When should it be used, and when should it not be used?

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3. What is Content Analysis in requirement modeling for WebApps? Explain it.

4. What factors should a Software Project Manager consider when choosing a Software Project Team?

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5. List four key elements of Requirement Analysis.

6. Briefly explain the CRC (Class-Responsibility-Collaborator) Model.

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7. Draw the symbols used in Class diagrams and label their names.

Section B - Case Study (11 X 1 = 11 Marks)

Case Study: Modernizing an Airline Ticket Booking System (ATBS)

Introduction: Airjet Airlines, a well-established global airline, is facing challenges with its outdated Airline Ticket Booking System (ATBS). To keep up with industry demands, improve customer satisfaction, and streamline operations, the airline has decided to modernize its ticket booking system. The new system aims to provide a seamless user experience, enhance security, and be scalable for future growth.

System Overview: The system should allow customers to search for flights based on destinations, travel dates, and class preferences (economy, business, first-class). Once a flight is selected, customers can book tickets, choose their seats, and make payments online through multiple payment options. The system must also provide customers with an option to view, modify, or cancel bookings and issue electronic tickets and boarding passes.

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Frequent flyers should be able to log in to their accounts to access their loyalty points, redeem them for rewards, and track their travel history. Customers should receive notifications for flight status changes, cancellations, or gate changes via email or SMS.

The system must support integration with third-party applications such as payment gateways, loyalty program providers, and airport services. It should be accessible on multiple devices, including desktops, tablets, and mobile phones, with a user-friendly interface.

List out the functional and non-functional requirements. 4 Marks

Draw a use case diagram of the above-given problem. 4 Marks

Write use case description of use case Book Ticket . 2 Marks

Draw activity diagram of Book Ticket use case 1 Marks

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